

Colin Gregg

Irvine, California | 603-801-6620 | colingregg03@gmail.com | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Lehigh University, Integrated Business and Engineering Honors Program

Graduating: May 2026

Degrees: Dual BS in Electrical Engineering and BS Finance

GPA: 3.8 / 4.0

Coursework: Electronic Circuits Design & Lab, Microcontrollers Lab, Power Electronics, Digital Signal Processing.

Honors: Tau Beta Pi Engineering Honor Society Member, Dean's List all semesters.

SKILLS

Hardware Design: KiCad, LTSpice, Synopsys, Cadence Virtuoso.

Lab & Testing: Oscilloscope, DMM, Function Generator, LCR Meter, Component Soldering, PCB Troubleshooting.

Embedded & Software: Tiva MCU, C/C++, SystemVerilog, MATLAB, Python.

Protocols and Components: SPI, UART, ADC/DAC Integration.

Business & Analytics: SAP ERP, Financial Modeling, BOM Optimization.

PROJECT EXPERIENCE

Underwater Direction of Arrival System (Capstone) – TX Power Amplifier Hardware Lead

- Engineered a Class D Push-Pull Power Amplifier driving an acoustic transducer to test an autonomous drone's navigation system by designing a 4-layer PCB in KiCad, isolating 3.3V digital and 200V/1.5A analog domains to prevent interference.
- Optimized high-current switching loops to reduce parasitic inductance by implementing wide polygon pours, an LC-pi filter, and tight node placement around the transformer and MOSFET drains.
- Maximized power transfer and minimized distortion to a piezoelectric transducer by designing custom LC impedance matching networks, utilizing an LCR meter to extract physical parasitics and validating mathematical models in LTSpice.

Underwater DoA System (Capstone) – RX LNA & Analog Edge Detector Hardware Lead

- Designed a four-channel analog front-end utilizing an INA217 instrumentation amplifier and an SPI-controlled VCA810 for continuous-time gain adjustment and flat group delay to implement a phase-based direction finding algorithm.
- Eliminated spurious oscillations on LNA boards by performing component-level troubleshooting on high-Q 2nd and 8th order bandpass filters and recalculating feedback networks and phase margins.
- Prevented a timely PCB respin by diagnosing an undocumented SPI-to-shift-register race condition on inherited hardware causing erratic DAC behavior by developing an STM32 firmware workaround to correct latch timing.
- Overcame a 200 μ s software delay for edge detection in a highly noisy environment by designing an analog edge detector to discern large transient noise from a transmitted signal using a precision envelope detector and tuned charging capacitor.

Discrete Operational Amplifier (Independent Project)

- Engineered a high-linearity discrete BJT op amp achieving a 2.3 MHz GBWP, 45dB open-loop gain, and <5% THD by designing and simulating a folded cascode Voltage Amplifier Stage (VAS) and Class AB output architecture in LTSpice.
- Validated hardware performance against theoretical models by evaluating a breadboard prototype with an oscilloscope to discover phase margin instability, leading to an iterative PCB layout redesign to reduce parasitics.

WORK EXPERIENCE

Schott Advanced Optics – Consultant

Duryc, PA, June 2025 – August 2025

- Developed an automated SAP purchase requisition system and interactive image-based machine drawing database, reducing machine down-time, optimizing supply chain efficiency, and reducing potential holding costs.

Batteryze – Business Development Intern

San Francisco, CA, May 2024 – August 2024

- Synthesized market research on EV batteries into investor pitch decks, developing strong cross-functional communication and technical translation skills.

Safran Optics 1 – Quality Engineering Intern

Bedford, NH, June 2023 – August 2023

- Participated in root-cause analysis meetings for thermal imager manufacturing defects, identifying hardware failures related to soldering and environmental contamination.